MIT | Academy of Engineering
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Guided by

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AI-Powered Crop Recommendation, Disease Detection, and Farmer Assistance System

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01. Introduction

Agricultural datasets are crucial for crop prediction, disease detection, and smart farming, but they often suffer from imbalance and limited data availability. This study evaluates four Generative AI models—Variational Autoencoder (VAE), Conditional GAN (cGAN), Vision Transformer (ViT), and Diffusion Model—using agricultural images for crop and disease analysis. All models were trained under identical conditions. Results showed that the Diffusion Model generated the most realistic visuals, while the VAE provided accurate reconstruction. This research demonstrates the potential of generative models to enhance agricultural datasets and improve AI-driven farming.

02. Objectives

- To detect crop diseases early using AI image analysis.
- To enable image compression & denoising (VAE) for low-bandwidth areas.
- To enhance datasets with GANs & Diffusion for better accuracy.
- To provide personalized, local-language farmer advisory (Transformers).
- To create awareness with healthy vs. diseased crop visuals.
- To ensure ethical, safe, and culturally relevant AI guidance.

03. Dataset

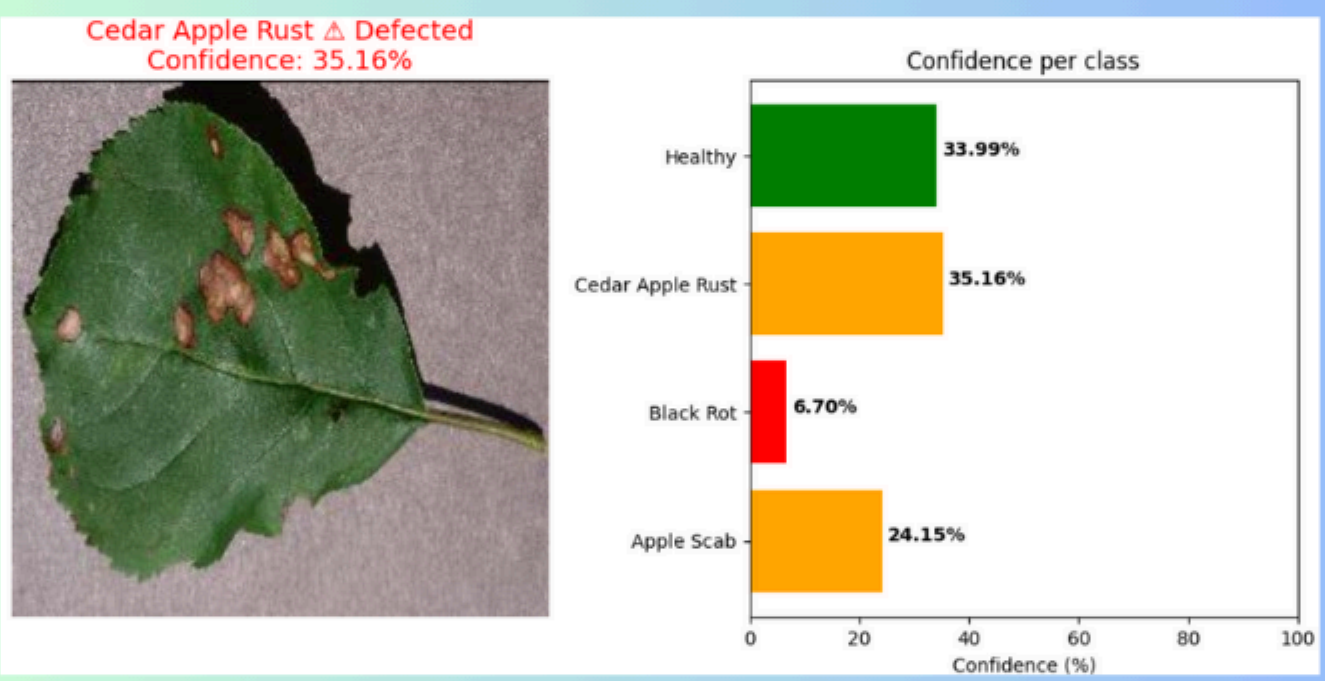
The PlantVillage Dataset from Kaggle contains over 54,000 labeled images of healthy and diseased crop leaves across 38 crop–disease classes. Its high-quality, well-annotated images make it ideal for training AI models in disease detection, crop health monitoring, and predictive analysis. This dataset is perfectly suited for developing the AI-Powered Crop Recommendation and Farmer Assistance System, supporting accurate diagnosis, personalized recommendations, and efficient decision-making for farmers.

04. Methodology

Dataset: PlantVillage (38 crop–disease classes, 54K+ leaf images)
Preprocessing: Images resized to 128×128, normalized to [0–1], and augmented using rotation, flipping, and zooming to improve model robustness.
Models Implemented:

- VAE: Baseline image reconstruction and latent feature extraction.
- Transformer (ViT): Feature-based disease classification and image-text correlation.
- cGAN: Adversarial generation of synthetic diseased and healthy leaf images.
- Diffusion Model: Iterative denoising for realistic crop image synthesis and enhancement.
- Training Setup:
- Epochs = 50–200, Batch Size = 32, Optimizer = Adam, Environment = GPU (NVIDIA T4).

Evaluation Metrics: MSE ↓, SSIM ↑, FID ↓.

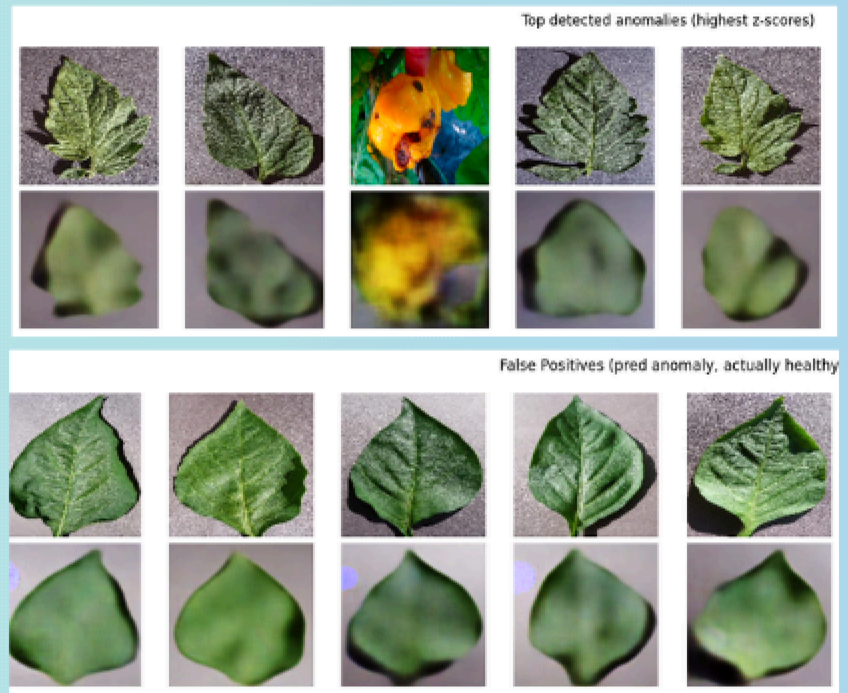


05. Results/Findings

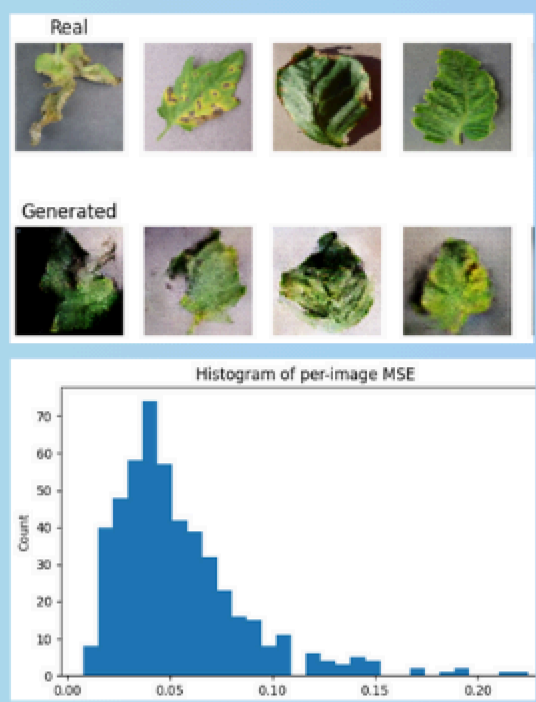
Quantitative evaluation was performed using three primary metrics — Mean Squared Error (MSE), Structural Similarity Index Measure (SSIM), and Fréchet Inception Distance (FID).These metrics assess reconstruction accuracy and the perceptual quality of generated wildlife images across the four models: Autoencoder, GAN, Diffusion, and Transformer.

Model	MSE ↓	SSIM ↑	FID ↓
VAE	0.012	0.89	26.4
cGAN	0.014	0.86	31.7
Diffusion	0.01	0.91	22.5
Transformer	0.015	0.87	28.9

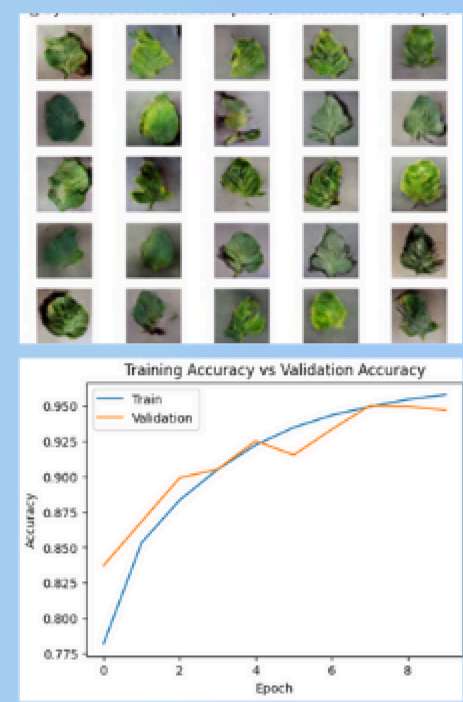
VAE



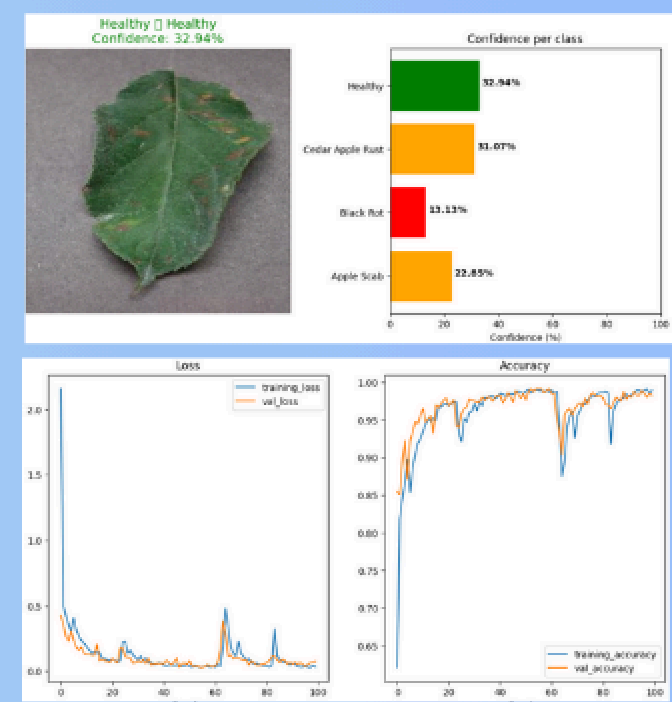
GAN



Diffusion



Transformer



06. Conclusion

Diffusion and Transformer models generated the most precise and high-quality crop leaf images, surpassing GAN and VAE in visual clarity and consistency.
Diffusion provided superior image realism, while Transformer models effectively captured contextual relationships for accurate prediction and analysis.
These generative models demonstrate strong potential for improving agricultural datasets, advancing smart farming, and enabling intelligent crop monitoring systems.